

1. What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

As a PhD nurse researcher (>25+ years) who has focused on making EHRs useful and usable to nurses in support of continuously improving EHRs and enhancing patient outcomes –I find the EHR vendors the biggest barrier to efforts of innovators to integrate AI into daily tools for nurses. Specifically, the big vendors (e.g., Epic) –make it difficult for innovators to easily, simply, and cost effectively help reduce the documentation burden and the effectiveness of clinicians. Access and integration are cleverly blocked by vendors –who claim – “we can do this” and through their efforts in adding band-aids to existing EHRs increase the complexity, cost, and documentation time. Further the new additions to legacy EHRs are not adequately evaluated for the overall impact on clinician’s workflow and patient outcomes.

2. What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? What HHS regulations, policies, or programs could be revisited to augment your ability to develop or use AI in clinical care? Please provide specific changes and applicable Code of Federal Regulations citations.

Many changes recommended by the HTI-5 proposed rule will provide some relief from the barriers EHR vendors employ to block superior innovations from the field from getting traction . It is still possible for vendors (deep pockets) to cleverly put up barriers that make it impossible for creative and knowledgeable innovations to see the light of day. I say this from my personal experiences – with one example being that my team created the nurse HANDS care planning software that was fully tested in multiple hospitals and demonstrated the simple ability to collect interoperable standardized nursing data in the same way across hospital systems with different vendors (much to the surprise of many). HANDS provided many ways to support the nurse to create and continuously update their nursing care plans with plenty of aids to support easy use. The hospitals wanted to adopt HANDS but big vendors said they could do what we had done – EPIC did a simple copy but their care planning system never became anything close to what the HANDS systems provided. In fact –the system as it exists in EPIC is not useful to the nurse. Try as we did –the vendor was successful in blocking our efforts. The enormous expertise of our research team far exceeded the vendor experiences including the absence of sufficient knowledge of nurses hired onto EPICs. Epic hired nurses to sell and advocate for the EPIC system and these nurses did not have adequate knowledge of EHRs to recognize the differences between HANDS and Epic and further were hired solely to advocate for EPIC. Recommendation – seek to find those pockets of experts (deep knowledge of the user group needs, AI, and technology) like our highly qualified team at the University of Florida College of Nursing.

3. For non-medical devices, we understand that use of AI in clinical care may raise novel legal and implementation issues that challenge existing governance and accountability structures (e.g., relating to liability, indemnification, privacy, and security). What novel legal and implementation issues exist and what role, if any, should HHS play to help address them?

Problem: LLM prompts can generate responses for which the source/s of evidence are not always clear. If adopted—without validating the evidence this could result in errors or adverse outcomes for patients. Overregulation of AI in the early phase however can inappropriately limit the use of AI which in turn denies the enormous clinical benefits that we see being generated by use. Further clinicians and patients are and will continue to use AI (LLMs) in a variety of ways in spite of regulation to augment their decision making – Important to find ways to make this easier and ensure safety.

Solution: Not all prompt responses by LLMs are necessarily capable yet of quickly providing the evidence that backs the response. Some of responses are so straightforward and face valid as to not need evidence for the clinician to weigh the value of suggestion for the patient. Further over regulation at this point could inappropriately slow use of AI in ways that patients and clinicians are unable to garner the tremendous value of LLMs. Ensuring safe and effective use is key here....in the early phases –it might be helpful for vendors to merely provide compelling rationale for why their AI powered products are safe rather than for government to mandate the creation of regulations for AI use that will be outdated within the year. Our bottom line suggestions – When using LLMs to generate clinical decision support (CDS) 1) ensure human is in the loop who decides to use/not use the CDS suggestions and is thus responsible for all decisions (industry not liable) 2) provide a disclaimer for CDS suggestions saying that clinician is expected to use one's critical thinking/clinical judgement in applying any suggestion and to seek further collaborating evidence when in doubt. For AI in prediction models/algorithms vendor must provide evidence of the how algorithms are computed and validated. Finally in general the government should consider creating principles that are helpful in guiding innovators to demonstrate the safety, strengths and limitations without resorting to stringent rules and regulations.

4. For non-medical devices, what are the most promising AI evaluation methods (pre- and post-deployment), metrics, robustness testing, and other workflow and human-centered evaluation methods for clinical care? Should HHS further support these processes? If so, which mechanisms would be most impactful (e.g., contracts, grants, cooperative agreements, and/or prize competitions)?

Our team strongly encourages the use of simulation in a representative set of use cases for which the AI (LLM) output can be evaluated for suitability against a gold standard for fit by a team of experts prior to implementation in practice. This procedure should be

described fully by the vendor to help the potential buyer/user make decisions about use based on their understanding of the validity and safety of the AI augmented predictive models and LLM generated CDS suggestions. Our team has done this when reviewing nursing care plan suggestions generated by Chat-GPT. We have also used RAGs to constrain automatic coding of the nursing diagnoses, interventions and outcomes to the latest version of the standardized terminologies used. Our testing has resulted in eliminating nearly 100% of fully inappropriate suggestions. We have also noticed that many of the LLMs now possess similar capability and might be used as a way to cross check a response with one or more of the LLMs to further validate the quality of the output (e.g., prompt is directed to multiple LLM models and outputs are compared).

5. How can HHS best support private sector activities (e.g., accreditation, certification, industry-driven testing, and credentialing) to promote innovative and effective AI use in clinical care?

AI capability is growing so rapidly that setting up rigorously developed and detailed guidelines for accreditation, certification, industry-driven testing, and credentialing is likely premature and if this approach is adopted progress will be substantially impeded. Further, traditional ways of managing transformational tools will not work well with AI and LLMs. I strongly recommend that the federal government seek to create contracts with truly qualified experts to develop, test, and continue to improve the safe and effective use of AI with a defined population. For example, the health care field to date has typically focused on applying AI to medicine and then adapting use for the other health care disciplines. This is costly and a waste of time. Nursing work and work flow differ dramatically from medical care and thus adapting medical forms of AI to nursing is wrong – minded and has created tools that work very poorly for nursing. Such tools end up costing more in terms of money spent on the product, time spent trying to use the product that does not work well, and the errors that result from the poor quality tool. Further, most nurses do not have the knowledge and skill required to create safe and effective tools for nursing and yet get hired and paid well to market poor tools (e.g., lots of nurses employed by EPIC –who simply carry out the EPIC way without the skills and knowledge to see and question the gaps that exist in that product. Since nurses front for the product—it is unfairly assumed (in name) that the product is the best it can be for nurses. Again – our nursing focused team at UF comprises a phenomenal group of highly qualified nursing informaticists, engineers and technologists who possess the full knowledge, experience and commitment to quickly transform the way nurses use EHRs and AI in ways that reduce cost, increase care quality and patient and nurse satisfaction.

6. Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI

tools would have the greatest potential to improve health care outcomes, give new insights on quality, and help reduce costs?

I don't see tools that are actually working well in practice for nurse –because the giants in the industry block that progress by not allowing the competition of new start-ups led by creative groups to enter. Further the strategies the giants use to introduce new technology are band-aid related and do not look at the holistic impact of the new bell or whistle added to the already over complex and poorly connected technical systems. Most importantly the strategies used to test products are poor and not focused on creating universally useful products. Vendors hide behind the unique desires of their individual customers when in fact if a tool is designed well for practice it likely will work well across settings (be universally useful and interoperable). Tailoring products to the customer nuances is merely an excuse to justify making more money and ensuring interoperability never transpires. . Allowing this strategy to continue has resulted in little to now progress, increased cost, and increased complexity –accompanied by dramatically reduced quality.

7. Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

The IT administrators seem to hold the most power in the AI/IT decisions (administrators defer to them) and unfortunately these folks often do not hold sufficient clinical background nor AI knowledge to be afforded this high status. Further these IT officers become puppets of their EHR systems (e.g., EPIC) and tend to support blocking of key information to potential innovators – holding the party line—which is “our vendor will make it happen!” One need only look at the progress made in the past 20 years to know that the vendors are not making it happen. Instead their approaches are increasing costs and complexity without a increasing quality. This must stop and a focus on seeking ways to support innovation in the EHR space that is external to existing EHRs. We need tools that are universally (and legitimately) useful and interoperable for all clinicians. As I mentioned, our team created an outstanding care planning system that nurses used effectively in the exact same way in four different institutions (folks said this was not possible). It is the vendors who perpetuated the myth that EHRs must be fully tailored to support the uniqueness of each of their institutional clients. Instead of relying on IT folks who have been coopted by their industry EHR partners (e.g., EPIC) – perpetuating little to no cost effective improvements in our EHRs – the market must be opened to the creativity of individual and small businesses external to the current monopolistic EHR industry to enable the creation, testing, and implementation of AI enabled technology solutions that can quickly transform the stagnation that exists today. .

8. Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Please consider specific data types, data standards, and benchmarking tools.

A big gap in our health care system is the absence of access to interoperable nursing data that can be used to evaluate and improve health care. This is somewhat baffling given patients go to hospitals to receive nursing care and thus the quality and intensity of that care have a crucial impact on patient outcomes. While attempts have been made over the years, it has been difficult to gain the attention and resources needed to capture nursing data and use it to improve practice. The wherewithal (technology, AI etc) to collect and utilize nursing data cost effectively exists with groups IT nursing reearchers/entrepreneurs at University of Florida, College of Nursing but resources are needed to bring our work/tools to life in the clinical settings across the country and world.

9. What challenges within health care do patients and caregivers wish to see addressed by the adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

The challenges our team sees for patients, caregivers, and clinicians alike is that most do not recognize or understand the extraordinary advancements that can be made with AI when used simply and smartly. Thus education and hands-on use are needed to help the stakeholders see the many benefits of AI and through this experience also learn how to use AI safely.

10. Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

HHS needs to prioritize research in which AI is used as part of an overall approach to transforming health care. For example, a limited AI tool may work well in some areas and not in others, thus we recommend that the HHS prioritize funding of projects that utilize and fully test AI as part of a holistic approach (comprehensive approach) designed to dramatically improve documentation, communication, quality of care, and user satisfaction while reducing the cost. Open the market fully to enable innovators and start-up to transform the stagnation - the capability exists but it must be unleashed.

a. Are there published findings about the impact of adopted AI tools and their use clinical care?

Yes but these remain limited –continue to foster ways for those using and testing AI to get their methods and result published /available to the public quickly (incentivize this work and speed dissemination).

b. How does the literature approach the costs, benefits, and transfers of using AI as part of clinical care?

To date the literature has predominantly included studies that show a positive/negative impact of a narrowly focused feature/tool/AI on the health or care of the patient. Often times the cost in dollars if mentioned do not include the overall cost to the entire system. For example, ambient listening makes it easier to capture the full patient perspective in real time but the clinician is expected to later re-read and validate the summary of the experience and validate the output. This may or may not improve accuracy but clearly increases documentation time of the clinician which is already at 30-40% of the nurses time. In fact nursing documentation in US hospitals today costs >\$60B a year most of which brings little value at the point of care. Bottom line – it is crucial to find ways to carefully but simply evaluate the overall benefits/negative impact of using AI anywhere in the system. We support encouraging a report of the costs, benefits, and transfers to the overall system for using AI in any part of the system.

II. Paperwork Reduction Act

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